

		Baseline				Hypermutation				Strong ILS				Outgroup clade				Purifying				runtime	
		(1% ILS)				(1% ILS)				(89% ILS)				(1% ILS)				(0% ILS)					
		#sites	10k	1k	4k	244k	4k	7k	3k	160k	35k	800	10k	140k	30k	897	9k	155k	12k	1k	5k		1M
true ARG (oracle)	n _{out} = 0		0.61	0.74	0.56	0.50	0.59	0.76	0.80	0.50	0.60	0.64	0.51	0.50	0.60	0.68	0.51	0.50	0.61	0.74	0.55	0.50	2.3m
	n _{out} = 1		1.00	0.98	0.99	0.50	1.00	0.97	0.99	0.50	0.71	0.72	0.67	0.50	1.00	0.89	0.99	0.50	1.00	0.97	1.00	0.50	2.1m
	n _{out} = 3		1.00	0.99	1.00	0.69	1.00	0.98	0.99	0.67	0.99	0.95	0.98	0.68	1.00	0.95	1.00	0.95	1.00	0.99	1.00	0.70	2.7m
	n _{out} = 10		1.00	1.00	1.00	0.91	1.00	0.99	0.99	0.87	0.99	0.98	0.98	0.87	1.00	0.95	1.00	0.95	1.00	0.99	1.00	0.91	2.9m
max-parsimony (μ→0)	n _{out} = 0		0.61	0.74	0.56	0.50	0.59	0.76	0.80	0.50	0.60	0.64	0.51	0.50	0.60	0.68	0.51	0.50	0.61	0.74	0.55	0.50	2.1m
	n _{out} = 1		1.00	0.98	0.99	0.50	1.00	0.97	0.99	0.50	0.71	0.72	0.67	0.50	1.00	0.89	0.99	0.50	1.00	0.97	1.00	0.50	2.3m
	n _{out} = 3		1.00	0.99	1.00	0.69	1.00	0.98	0.99	0.67	0.99	0.95	0.98	0.68	1.00	0.95	1.00	0.95	1.00	0.99	1.00	0.70	2.5m
	n _{out} = 10		1.00	1.00	1.00	0.91	1.00	0.99	0.99	0.87	0.99	0.98	0.98	0.87	1.00	0.95	1.00	0.95	1.00	0.99	1.00	0.91	3.0m
local-tree mode	n _{out} = 0		0.59	0.73	0.56	0.50	0.59	0.75	0.80	0.50	0.59	0.63	0.51	0.50	0.59	0.67	0.51	0.50	0.58	0.71	0.55	0.50	4.5m
	n _{out} = 1		0.99	0.96	0.97	0.50	0.99	0.96	0.98	0.50	0.68	0.69	0.63	0.50	0.99	0.82	0.98	0.50	1.00	0.91	0.96	0.50	6.0m
	n _{out} = 3		1.00	0.97	0.99	0.69	1.00	0.97	0.98	0.66	0.96	0.92	0.94	0.67	1.00	0.86	0.99	0.95	1.00	0.96	0.98	0.70	5.0m
	n _{out} = 10		1.00	0.98	0.99	0.87	1.00	0.98	0.99	0.85	0.97	0.95	0.96	0.82	0.99	0.88	0.99	0.95	1.00	0.95	0.98	0.87	7.1m
SINGER	n _{out} = 0		0.54	0.69	0.56	0.50	0.56	0.73	0.81	0.50	0.54	0.61	0.51	0.50	0.54	0.64	0.51	0.50	0.54	0.68	0.55	0.50	2.2d
	n _{out} = 1		0.99	0.95	0.97	0.50	1.00	0.96	0.98	0.50	0.69	0.71	0.65	0.50	0.99	0.80	0.98	0.50	1.00	0.92	0.97	0.50	2.9d
	n _{out} = 3		1.00	0.93	0.98	0.69	1.00	0.96	0.97	0.66	0.94	0.90	0.92	0.67	1.00	0.85	0.99	0.94	1.00	0.94	0.98	0.70	4.9d
	n _{out} = 10		1.00	0.90	0.97	0.86	1.00	0.96	0.97	0.85	0.95	0.91	0.93	0.75	1.00	0.86	0.99	0.95	1.00	0.93	0.98	0.78	10.3d
SINGER (fixed-tree)	n _{out} = 0		0.53	0.69	0.56	0.50	0.54	0.72	0.81	0.50	0.53	0.60	0.51	0.50	0.53	0.64	0.51	0.50	0.53	0.67	0.55	0.50	2.2d
	n _{out} = 1		0.99	0.95	0.98	0.50	1.00	0.96	0.98	0.50	0.71	0.72	0.68	0.50	1.00	0.80	0.99	0.50	1.00	0.92	0.97	0.50	2.9d
	n _{out} = 3		1.00	0.92	0.97	0.61	1.00	0.96	0.97	0.66	0.93	0.89	0.90	0.67	1.00	0.86	0.99	0.94	1.00	0.93	0.98	0.70	4.9d
	n _{out} = 10		1.00	0.91	0.97	0.72	1.00	0.96	0.96	0.87	0.94	0.91	0.92	0.79	1.00	0.85	0.99	0.95	1.00	0.93	0.97	0.79	10.6d
tsinfer + tsdate	n _{out} = 0		0.51	0.67	0.56	0.50	0.51	0.70	0.81	0.50	0.50	0.58	0.51	0.50	0.50	0.60	0.51	0.50	0.50	0.65	0.55	0.50	26.6m
	n _{out} = 1		0.97	0.95	0.94	0.50	0.99	0.95	0.98	0.50	0.72	0.70	0.71	0.50	0.98	0.80	0.93	0.50	0.95	0.93	0.86	0.50	26.6m
	n _{out} = 3		0.96	0.92	0.93	0.67	1.00	0.95	0.96	0.65	0.78	0.77	0.75	0.59	0.98	0.84	0.93	0.94	0.97	0.94	0.92	0.70	28.7m
	n _{out} = 10		0.98	0.83	0.91	0.91	0.99	0.94	0.94	0.86	0.84	0.83	0.79	0.81	0.98	0.79	0.92	0.93	0.97	0.83	0.90	0.91	31.2m
Relate	n _{out} = 0		0.51	0.68	0.56	0.50	0.52	0.71	0.81	0.50	0.51	0.57	0.51	0.50	0.51	0.62	0.51	0.50	0.52	0.67	0.55	0.50	5.1m
	n _{out} = 1		0.99	0.95	0.98	0.50	1.00	0.95	0.97	0.50	0.72	0.70	0.72	0.50	0.99	0.80	0.98	0.50	1.00	0.90	0.96	0.50	5.4m
	n _{out} = 3		0.99	0.94	0.98	0.69	1.00	0.95	0.96	0.66	0.85	0.83	0.80	0.67	1.00	0.83	0.98	0.93	1.00	0.93	0.98	0.70	6.7m
	n _{out} = 10		0.99	0.90	0.96	0.91	1.00	0.95	0.95	0.85	0.88	0.84	0.83	0.84	1.00	0.80	0.98	0.93	1.00	0.88	0.97	0.91	10.0m
fixed-tree mode	n _{out} = 0		0.50	0.67	0.56	0.50	0.50	0.69	0.81	0.50	0.50	0.57	0.51	0.50	0.50	0.60	0.51	0.50	0.50	0.64	0.55	0.50	2.3m
	n _{out} = 1		1.00	0.95	0.99	0.53	1.00	0.93	0.94	0.52	0.72	0.70	0.73	0.50	1.00	0.79	0.99	0.70	1.00	0.90	0.96	0.54	6.8m
	n _{out} = 3		1.00	0.97	0.99	0.69	1.00	0.95	0.94	0.67	0.77	0.77	0.74	0.67	1.00	0.85	0.99	0.93	1.00	0.95	0.98	0.70	7.1m
	n _{out} = 10		1.00	0.98	0.99	0.91	1.00	0.96	0.95	0.87	0.77	0.78	0.70	0.85	1.00	0.82	0.99	0.92	1.00	0.95	0.98	0.91	58.1m
fixed-tree (adaptive)	n _{out} = 0		0.50	0.67	0.56	0.50	0.50	0.69	0.81	0.50	0.50	0.57	0.51	0.50	0.50	0.60	0.51	0.50	0.50	0.64	0.55	0.50	2.4m
	n _{out} = 1		1.00	0.95	0.99	0.53	0.99	0.93	0.94	0.52	0.72	0.70	0.73	0.50	1.00	0.79	0.99	0.70	1.00	0.90	0.96	0.54	6.5m
	n _{out} = 3		1.00	0.97	0.99	0.69	1.00	0.95	0.95	0.67	0.77	0.77	0.74	0.67	1.00	0.85	0.99	0.93	1.00	0.95	0.98	0.70	8.3m
	n _{out} = 10		1.00	0.98	0.99	0.91	1.00	0.96	0.95	0.87	0.77	0.76	0.70	0.85	1.00	0.83	0.99	0.92	1.00	0.95	0.98	0.91	58.2m
majority outgroup	n _{out} = 0		0.50	0.67	0.56	0.50	0.50	0.69	0.81	0.50	0.50	0.57	0.51	0.50	0.50	0.60	0.51	0.50	0.50	0.64	0.55	0.50	1.3m
	n _{out} = 1		1.00	0.94	0.99	0.56	1.00	0.90	0.91	0.55	0.72	0.69	0.73	0.51	1.00	0.72	0.99	0.89	1.00	0.89	0.96	0.57	1.3m
	n _{out} = 3		1.00	0.91	0.97	0.69	1.00	0.91	0.88	0.66	0.95	0.89	0.93	0.68	1.00	0.81	0.99	0.93	1.00	0.93	0.97	0.70	1.4m
	n _{out} = 10		1.00	0.91	0.97	0.69	1.00	0.92	0.87	0.69	0.96	0.95	0.96	0.67	1.00	0.81	0.99	0.93	1.00	0.93	0.97	0.70	1.6m
			ISM bi	≥2 mut	i=n-1	fixed	ISM bi	≥2 mut	i=n-1	fixed	ISM bi	≥2 mut	i=n-1	fixed	ISM bi	≥2 mut	i=n-1	fixed	ISM bi	≥2 mut	i=n-1	fixed	